SideQuest:
Gamified Local Community & Gig Platform

Fundamentals of Software Engineering

Deliverable Number: Deliverable # 4 (Iteration 3 / Sprint 3)

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Group Members:

- Talmeez ur Rehman (Roll Number: 24L-2576) - Project Lead & Frontend UI/UX Developer
- Muhammad Yaseen (Roll Number: 24L-2515) - Backend Developer & Software Architect
- Taimur Amir (Roll Number: 24L-2518) - Requirement Engineer & QA Tester

Project Title and Introduction:

Project Title: SideQuest: Gamified Local Community & Gig Platform

Introduction: SideQuest is a hybrid community-building and gig-economy web application designed to gamify daily tasks and social needs into localized "Quests." While Sprint 2 introduced secure escrow wallets and mutual completions, Sprint 3 delivers the ultimate **End-Game Polish, Moderation, and Community Features**. This final iteration focuses on deploying a robust Tier 1 KYC Verification system, Grandmaster (Admin) Moderation tools (Ban Hammer & KYC Revocation), Quest-Isolated Tavern Chat (Direct Messaging), Custom Avatars, and a Global Hall of Fame to establish a thriving, trustworthy digital guild.

User Stories (Sprint Backlog):

US-08: KYC Verification & Admin Moderation

Story ID	US-08	**Story Title**	KYC Verification & Admin Moderation
Priority	Critical	**Estimated Hours**	10
User Story	**As an** adventurer, **I want** to securely upload my CNIC images **so that** I can earn a Tier 1 Verification badge and build trust. **As an** Admin, **I want** a dedicated dashboard **so that** I can approve, reject, or revoke KYC status for users.		
Description	This story introduces a multi-step verification system. Users upload compressed images of their Royal Seal (CNIC), entering a 'Pending' queue. A Grandmaster (Admin) reviews these images in a dedicated dashboard to approve or reject them, unlocking the user's ability to post bounties.		
Sub User Stories	• As a user, I want my images to be compressed client-side before upload to save bandwidth. • As an Admin, I want to view full-screen overlays of the submitted CNIC images. • As an Admin, I want the ability to revoke a previously granted KYC badge.		
Acceptance Criteria	• Only users with IsAdmin = 1 can access the /admin route.		

	• Non-verified users cannot post new bounties. • Image payloads must be securely parsed and stored as Base64 strings in the database.

US-09: Tavern Chat (Direct Messaging)

Story ID	US-09	**Story Title**	Tavern Chat (Direct Messaging)
Priority	High	**Estimated Hours**	8
User Story	**As a** hired adventurer or Quest Giver, **I want** to chat directly with my quest partner **so that** we can coordinate the details of the ongoing gig.		
Description	Once a quest hits the 'Accepted' or 'InProgress' phase, a secure, isolated chat room opens between the Quest Giver and the Quest Taker. The chat is strictly bound to the specific QuestID.		
Sub User Stories	• As a user, I want the chat to automatically fetch new messages without refreshing the page. • As a user, I want the chat to be disabled/hidden once the quest is marked 'Completed'.		
Acceptance Criteria	• Messages must be strictly isolated using a composite check of QuestID, SenderID, and ReceiverID. • Timestamps must parse UTC server time into the user's local PC timezone.		

US-10: Hall of Fame & Custom Avatars

Story ID	US-10	**Story Title**	Hall of Fame & Custom Avatars
Priority	Medium	**Estimated Hours**	6
User Story	**As a** user, **I want** to upload a custom avatar and view the top-rated adventurers on a Leaderboard **so that** I can customize my identity and strive for community recognition.		

Description	This story adds RPG flair to the platform. Users can replace the generic user icon with a custom portrait. A new "Hall of Fame" tab ranks the top 10 adventurers based on their 5-star ratings and total quests completed.
Sub User Stories	• As a user, I want to see avatars displayed on quest cards, the navbar, and in the chat room. • As a user, I want to see how many total reviews an adventurer has received.
Acceptance Criteria	• The Leaderboard must calculate the Top 10 users using an ORDER BY Rating DESC, QuestsCompleted DESC SQL query. • Avatars must gracefully fallback to a default <User /> icon if no image is uploaded.

US-11: Grandmaster Enforcement & Sidebar UI

Story ID	US-11	**Story Title**	Grandmaster Enforcement & UI Overhaul
Priority	High	**Estimated Hours**	7
User Story	**As an** Admin, **I want** to wield a 'Ban Hammer' and view platform statistics **so that** I can monitor the economy and suspend bad actors. **As a** user, **I want** a persistent Sidebar navigation **so that** I can switch pages easily.		
Description	Introduces global Admin statistics (Total Users, Gold in Economy, Completed Quests) and the ability to toggle account suspensions. Also overhauls the frontend UI, migrating from a cramped top-navbar to a sleek, responsive Sidebar dashboard.		
Sub User Stories	• As an Admin, I want to see a list of all Guild Members to suspend them. • As a user, I want a magical search bar to filter quests by title and location. • As a user, I want custom, themed modals to replace ugly native browser alert() popups.		

Acceptance Criteria	• A suspended user must be blocked at the /auth/login endpoint. • The Sidebar must snap to a bottom horizontal navigation bar on mobile devices. • Admins cannot suspend themselves.

Structured Specifications

US-08.1: KYC Document Compression & Upload

Type	Specification
Performance	The frontend must employ HTML5 Canvas to mathematically scale and compress image payloads to under 100KB before transmission to prevent SQL Server timeouts.
Functional	The system must parse and store the compressed image payload as a Base64 string within the `VARCHAR(MAX)` database columns.
UX	The UI must display an instant thumbnail preview of the selected CNIC documents before the user submits the form.

US-08.2: Grandmaster KYC Resolution & Revocation

Type	Specification
Security	The backend must implement a dual-layer middleware (`protect` and `adminOnly`) to restrict the resolution and revocation endpoints to users with `IsAdmin = 1`.
UX	The Admin Panel must render a full-screen, high-resolution overlay of the CNIC images upon clicking the thumbnails.
UX	Revoking a user's KYC must trigger a custom-styled React warning modal requiring explicit confirmation, bypassing native browser `alert()` or `confirm()` boxes.

US-09.1: Quest-Isolated Real-time Chat

Type	Specification
Functional	The backend must isolate message threads using a composite SQL filter requiring an exact match of `QuestID`, `SenderID`, and `ReceiverID`.
Performance	The frontend must implement a 3-second interval polling mechanism using `setInterval` to fetch new messages asynchronously without requiring a full DOM refresh.
UX	The chat interface must automatically scroll to the bottom (`bottomRef.current?.scrollIntoView()`) upon receiving or sending a new message.

US-09.2: Chat Lifecycle Management & Timezones

Type	Specification
Functional	The system must log message timestamps using `GETUTCDATE()` on the server and translate them to the user's localized PC timezone via JavaScript's `toLocaleTimeString()`.
UX	The "Send Direct Message" button must be dynamically hidden from the UI the moment a quest's status transitions to the 'Completed' phase.

US-10.1: Custom Avatars and Hall of Fame

Type	Specification
Functional	The Leaderboard backend must rank the top 10 users using a combined `ORDER BY Rating DESC, QuestsCompleted DESC` SQL query.
Functional	The frontend must safely decode SQL `BIT` fields to prevent Buffer serialization errors from falsely awarding KYC badges to unverified adventurers on the leaderboard.
UX	The UI must gracefully degrade to a default `<User/>` SVG icon if an adventurer has a `null` or empty `AvatarUrl` string.

US-11.1: Ban Hammer Enforcement

Type	Specification
Security	The authentication controller must check the `IsSuspended` BIT flag during login and return a 403 Forbidden status if it evaluates to true.
Functional	The backend must verify that the target user of a suspension does not possess `IsAdmin = 1` privileges to prevent Grandmasters from banning each other.
Functional	The `getGuildMembers` SQL query must explicitly exclude the currently logged-in Admin's ID so they cannot accidentally suspend themselves.

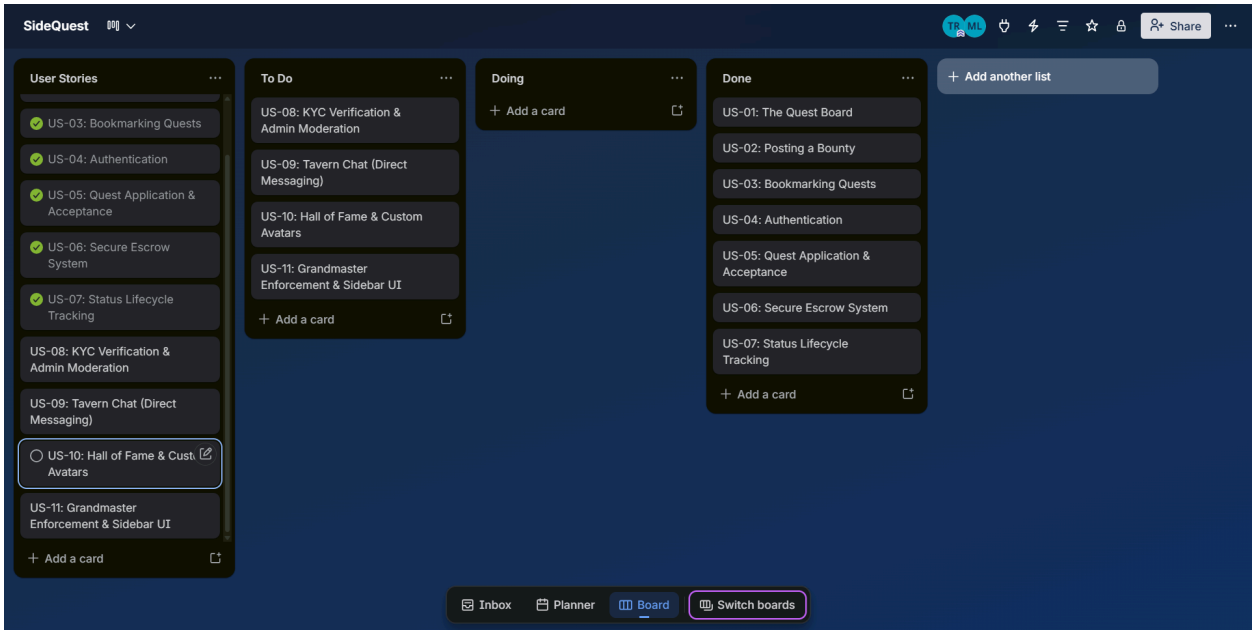
US-11.2: Persistent Sidebar Navigation & Search

Type	Specification
UX	The frontend architecture must deploy a fixed left-aligned Sidebar on desktop viewports that dynamically snaps to a horizontal, scrollable Bottom Navigation Bar on mobile devices.
Functional	The magical search input must execute a real-time `.filter()` function on the quest array, checking both `Title` and `Location` strings simultaneously.

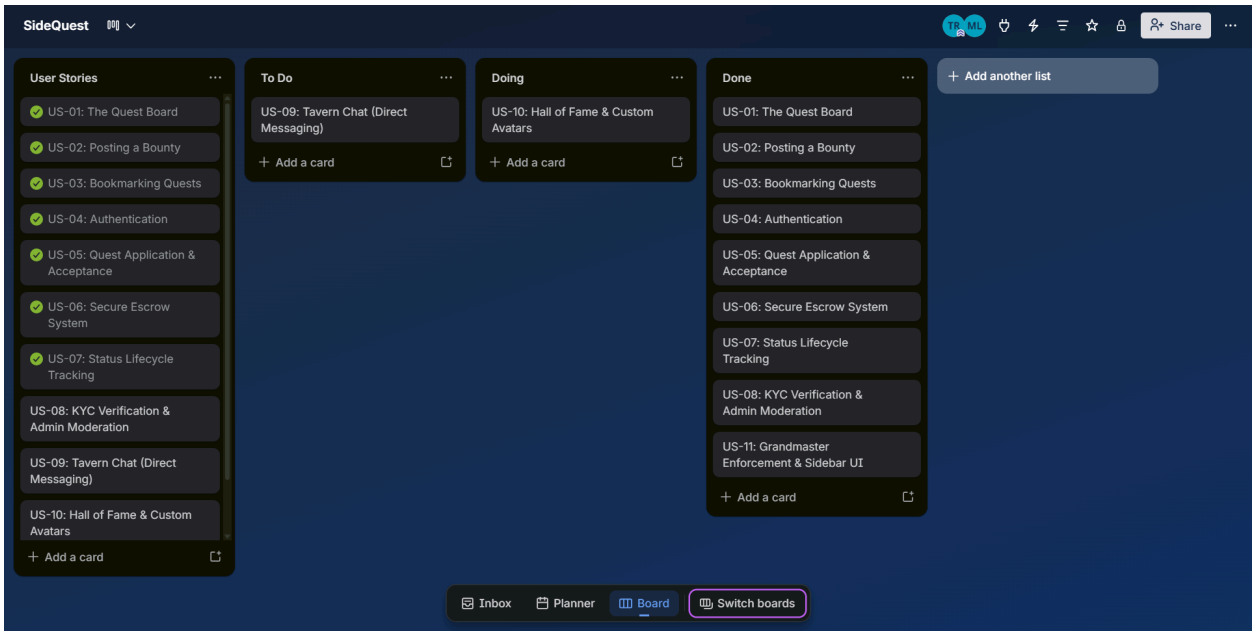
Scrum Board

(*Trello board link:* https://trello.com/invite/b/69bda26c79ab58133808b311/ATTI3b3a5945ffd93cb03da3050e104f2e035DD1600A/sidequest)

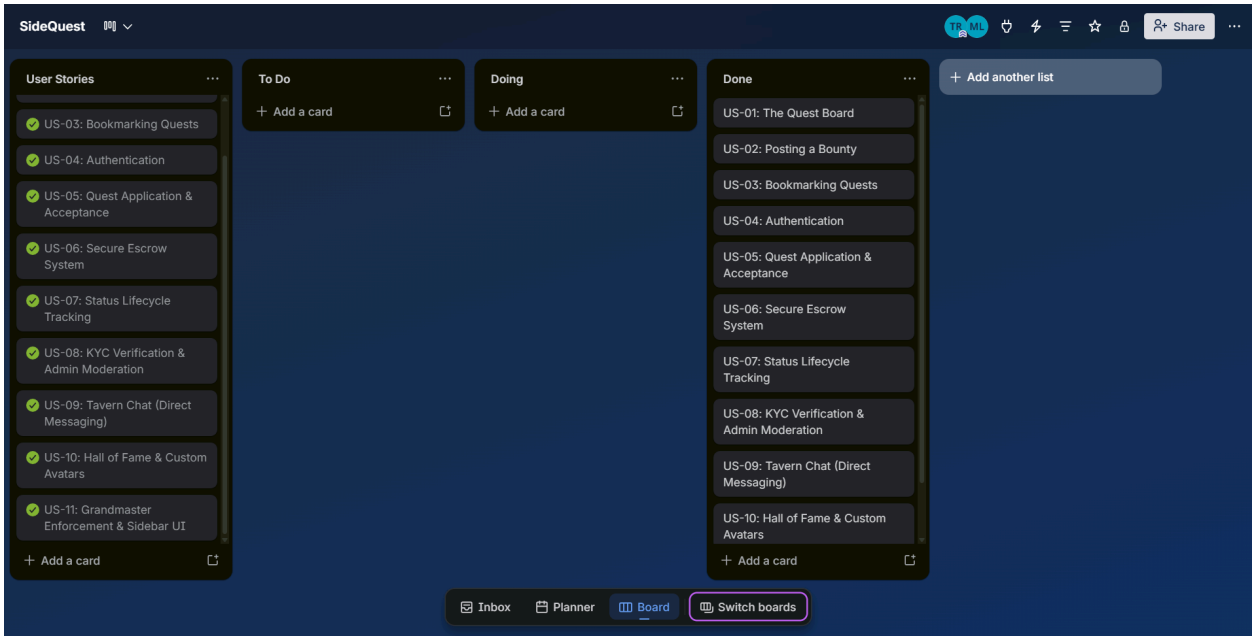
- **Snapshot 1 (Start of Sprint):**



- **Snapshot 2 (Middle of Sprint):**



- **Snapshot 3 (End of Sprint):**



- **Product Burn-Down Chart:**

NFR Specification:

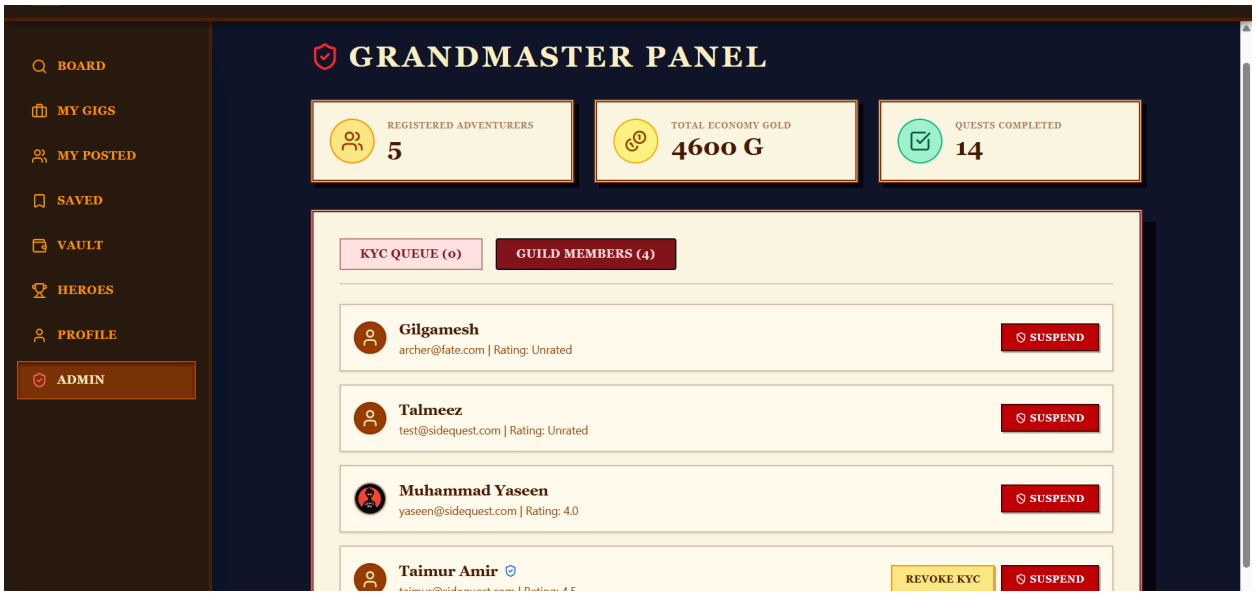
Performance (Image Optimization): To prevent ETIMEOUT crashes on the MSSQL database driver, all user-uploaded images (KYC and Avatars) must be strictly compressed client-side via HTML5 canvas algorithms before transmission, reducing 5MB payloads to <100KB strings.

Security (Role-Based Access Control): Admin-specific API endpoints (e.g., resolving KYC, viewing platform stats) must be strictly safeguarded by a dual-layer middleware system: validating the JWT token (protect), and verifying the user's explicit administrative privileges (adminOnly).

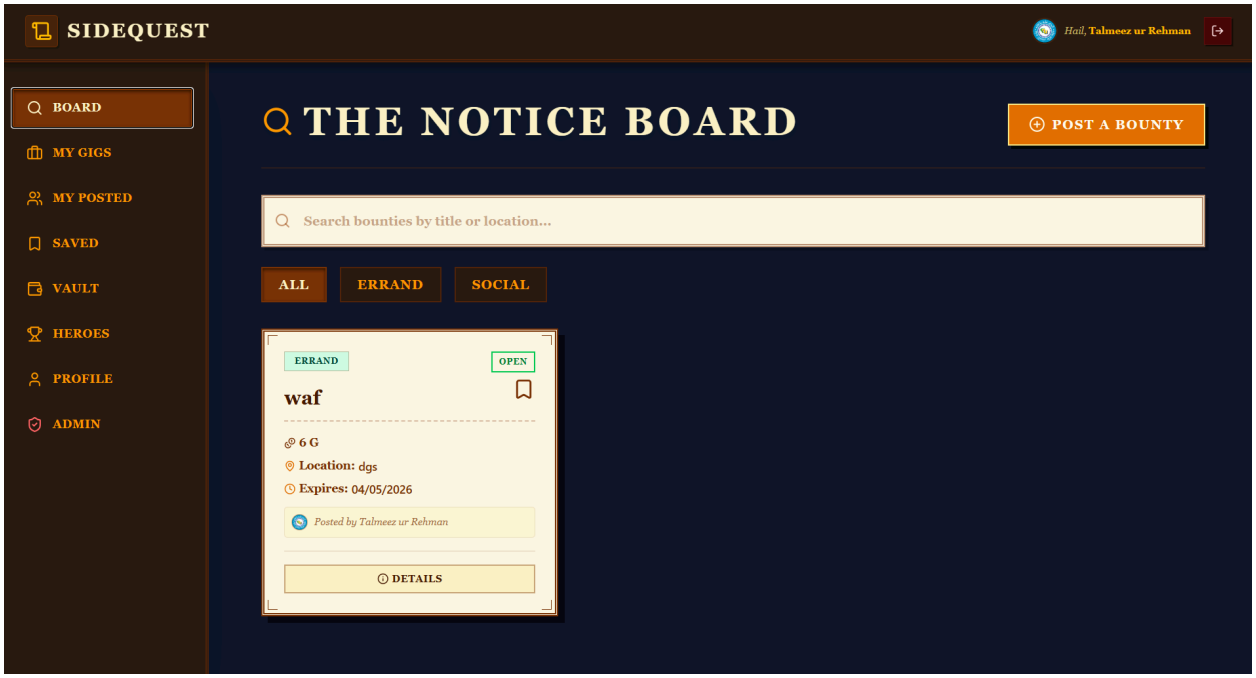
Usability (Non-blocking UI & Modals): The application must eliminate native, thread-blocking browser dialogue boxes (e.g., alert(), confirm()). All destructive actions (Revoking KYC, Cancelling Quests) must utilize custom, z-index managed React overlays to maintain seamless RPG immersion.

Iteration 3 Implementation Screenshots

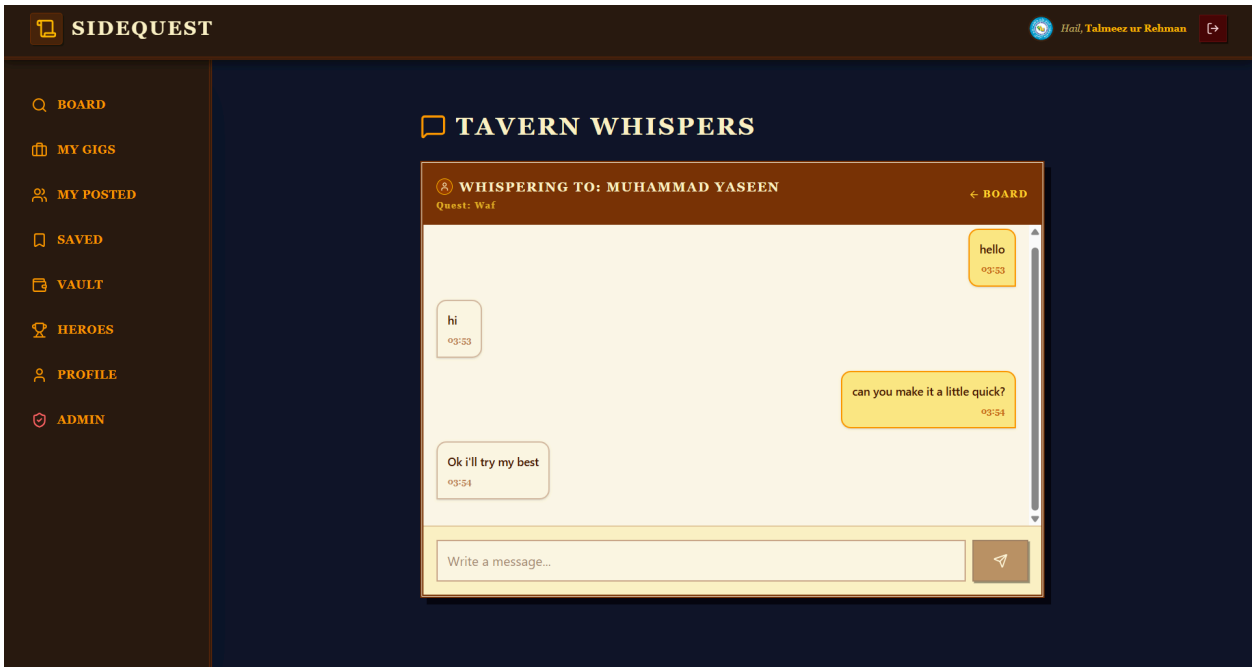
- **Frontend - The Grandmaster Admin Panel:**



- **Frontend - The New Sidebar Navigation & Magical Search:**



- **Frontend - Tavern Chat & Avatars:**



Work Division

Talmeez ur Rehman (Project Lead): Designed the image compression for Avatars and KYC. Engineered the massive UI layout shift to the responsive Sidebar (Dashboard) paradigm. Built the real-time Tavern Chat UI, the magical Search Bar filter logic, and completely replaced native

browser alerts with custom z-indexed React modal overlays.

Muhammad Yaseen: Developed the Admin/Grandmaster backend controllers and adminOnly security middleware. Patched the SQL database schema (expanding to VARCHAR(MAX)) to handle Base64 image strings safely. Engineered the quest-isolated Tavern Chat SQL queries with UTC timezone safety, and implemented the complex UNION queries for the Hall of Fame Leaderboard algorithms.

Taimur Amir: Managed the final iteration Trello backlog. Conducted rigorous QA edge-case testing (uncovering the SQL Buffer true/false illusion bug and the overlapping Quest Rating UI glitches). Documented the final Iteration 3 specifications and compiled the final comprehensive project reports.